

R_4) that is proportional to the voltage output of IC1. Electronic switch CD4066 modulates the current through the meter, causing variation of the duty cycle proportional to the voltage of the test power circuit.

The ramp duty-cycle generator consists of ramp voltage generator IC2(B), comparator IC2(C), timing oscillator NE555 and logic gates. The voltage of the test circuit is attenuated

and directly fed to the non-inverting input of IC2(C). NE555 is set to oscillate at a moderate frequency of around 800 Hz. The ramp needs periodic discharge of capacitor C2 for repetitive production of ramp profile. This is done by transistor T1.

For null offset, use a 10-kilo-ohm multiturn trimpot VR1 with 15-kilo-ohm resistors R4 and R5 on either side terminals of the wiper. Eliminate the

offset when power is not applied to the test circuit by measuring voltage V_i with a 0-200mV digital voltmeter. Adjust VR1 until the reading on the digital voltmeter is 0.00. It is easy to adjust calibrating resistor R_4 by measuring known power parameters like 5V and 400 mA. R_6 is the shunt resistor whose value is to be chosen such that the meter has a full-scale deflection for 5 mA. ●

Clap-Based Digital Volume Control

RAJ K. GORKHALI

This clap-based digital volume control offers four levels of volume control. With each clap, the volume of the audio system increases. After the fourth clap, the level of volume returns to the first or minimum level.

The circuit consists of a microphone, a level detector, a 4-stage counter and four analogue switches connected to a resistive ladder network.

The four outputs of the CD4040 counter are available at pins 9, 7, 6 and 2. These outputs are connected to the level detector formed by LED1 through

LED4 and control input pins 13, 5, 12 and 6 of IC CD4066 quad bilateral switch. Resistors R9 through R12 form the ladder network.

The circuit is powered by regulated 5V and consumes only a few milliamperes of current. The audio input signal from sources such as MP3 player and PC is fed to pin 8 of CD4066. The output of the circuit is taken from pin 1 of CD4066 and can be fed to any audio power amplifier.

The clapping signal captured by the electret microphone (MIC1) is amplified by transistor T1. MIC1 gets its bias voltage through resistor R1.

Counter IC CD4040 (IC1) gets a

clock pulse at its pin 10 when a certain noise level (threshold) is received through T1. The counter output determines the configuration of the four electronic switches inside IC2.

When output pin 9 of IC1 goes high, control pin 13 of IC2 shorts its pins 1 and 2, bypassing resistor R9 in the audio output line. Similarly, control pins 5, 12 and 6 of IC2 bypass resistors R10, R11 and R12, respectively. Thus, as per the sequential output across IC1, the series resistors are effectively introduced or removed in the audio output signal line. The maximum audio output is obtained when resistors R9 through R12 are shorted. The level of the audio signal can be seen on the LEDs (LED1 through LED4), with LED4 showing the maximum level.

Switch S1 allows the counter to reset, thus switching off all the switches of CD4066. In this case, attenuation in the audio output path is maximum as none of resistors R9 through R12 is shorted.

To calibrate the circuit, disconnect the clock input (pin 10) of CD4040 from the collector of T1. Now apply a pulse to the clock input by briefly connecting it to the +5V line. You will see the counter outputs change the state sequentially and so also the bilateral switches in CD4066. ●

